

Lecture 7

Inner Product Spaces

So far our vector spaces have had no geometry: no length, no angle, no notion of two vectors being perpendicular. The *inner product* supplies all three at once. You already know it on $\mathbb{R}^2$ and $\mathbb{R}^3$ from your mathematics course. Now we define it on any vector space, including a complex one.

The complex inner product is the foundation of quantum mechanics. In the Quantum Mechanics course you will learn that the state of a physical system is described by a complex vector, and that the inner product of two states gives the probability that a system in one state will be found in the other. We write the inner product in the *physics convention* and in the bra–ket notation physicists use, due to Dirac. The main result of the lecture, the Cauchy–Schwarz inequality, will show that such a probability never exceeds 1.

A word on convention. We take the inner product to be *conjugate-linear in the first slot and linear in the second*. This is the usual choice in physics: it makes $\langle \phi | \psi \rangle$ linear in the second vector, the ket $|\psi\rangle$. Many mathematics texts put the conjugation in the second slot; their formulas convert by swapping the two arguments.

Remark 7.1 (What Lectures 7–10 cover). This lecture and the next three work only with finite-dimensional spaces over $\mathbb{R}$ or $\mathbb{C}$. The physics examples use complex spaces, because quantum states are described by complex vectors. A vector and any nonzero multiple of it describe the same physical state. So we usually describe states by unit vectors (of length 1). Even then, the unit vectors v and $e^{i\theta}v$, with $\theta \in \mathbb{R}$, describe the same state.

Learning objectives

By the end of this lecture you will be able to:

- ▶ Verify the inner-product axioms in the physics convention (conjugate-linear first slot).
- ▶ Translate between the notation $\langle \cdot, \cdot \rangle$ and Dirac's notation $\langle \cdot | \cdot \rangle$.
- ▶ Compute norms, check orthogonality, and apply the Pythagorean theorem.
- ▶ Prove and apply the Cauchy–Schwarz inequality and the triangle inequality.
- ▶ Recover the inner product from the norm by the polarization identity.

1 Inner products

Definition 7.2 (Inner product, physics convention). An *inner product* on a complex vector space V assigns to each ordered pair $u, v \in V$ a scalar $\langle u, v \rangle \in \mathbb{C}$ such that

1. *conjugate symmetry*: $\langle u, v \rangle = \overline{\langle v, u \rangle}$;
2. *linearity in the second slot*: $\langle u, av + bw \rangle = a\langle u, v \rangle + b\langle u, w \rangle$;
3. *positive definiteness*: $\langle v, v \rangle \geq 0$, with equality iff $v = 0$.

A vector space with an inner product is an *inner product space*.

By conjugate symmetry $\langle v, v \rangle = \overline{\langle v, v \rangle}$, so the number $\langle v, v \rangle$ is real. Hence the positive-definiteness condition makes sense. Combining conjugate symmetry with linearity in the

second slot gives *conjugate-linearity in the first slot*:

$$\langle au + bw, v \rangle = \overline{\langle v, au + bw \rangle} = \overline{a\langle v, u \rangle + b\langle v, w \rangle} = \bar{a}\langle u, v \rangle + \bar{b}\langle w, v \rangle.$$

So scalars come out of the first slot conjugated and out of the second slot unchanged. Such a product is “one and a half” times linear, and is called *sesquilinear*. Over the reals the conjugation changes nothing. So a real inner product is a symmetric, positive-definite *bilinear* form, that is, one that is linear in each slot separately. Bilinear forms are studied in their own right in Lecture 11.

Example 7.3 (The standard inner products). On $\mathbb{C}^n$, the standard inner product is

$$\langle u, v \rangle = \sum_{k=1}^n \bar{u}_k v_k = u^\dagger v.$$

The coordinates of the *first* vector are conjugated, and $u^\dagger = \bar{u}^\top$ is the conjugate transpose (with u written as a column). On $\mathbb{R}^n$ the conjugation changes nothing, and we get the familiar dot product $\sum_k u_k v_k$. On the space of continuous functions on $[a, b]$,

$$\langle f, g \rangle = \int_a^b \overline{f(t)} g(t) dt$$

is an inner product. In quantum mechanics, the states of particles are described in similar function spaces. ◀

Example 7.4 (A computation in $\mathbb{C}^2$). With $u = (1, i)$ and $v = (2, 1 - i)$ in $\mathbb{C}^2$,

$$\langle u, v \rangle = \bar{1} \cdot 2 + \bar{i} \cdot (1 - i) = 2 + (-i)(1 - i) = 2 + (-i - 1) = 1 - i,$$

$$\langle v, u \rangle = \bar{2} \cdot 1 + \overline{(1 - i)} \cdot i = 2 + (1 + i)i = 2 + (i - 1) = 1 + i.$$

So $\langle v, u \rangle = \overline{\langle u, v \rangle}$, as conjugate symmetry requires. ◀

Example 7.5 (The Frobenius inner product). On $M_{n \times n}(\mathbb{C})$, the product

$$\langle A, B \rangle = \text{tr}(A^\dagger B) = \sum_{j,k=1}^n \bar{A}_{jk} B_{jk}$$

is an inner product. It is the standard inner product on $\mathbb{C}^{n^2}$ in disguise: write all n^2 entries of the matrix as one long vector. Later we will use it to measure the “size” of operators. ◀

Example 7.6 (A weighted product and a counterexample). With fixed weights $w_1, \dots, w_n > 0$,

$$\langle u, v \rangle = \sum_k w_k \bar{u}_k v_k$$

is also an inner product on $\mathbb{C}^n$. The condition $w_k > 0$ is needed for positive definiteness. For example, on $\mathbb{C}^2$ with weights $w = (1, -1)$, the nonzero vector $v = (0, 1)$ would have $\langle v, v \rangle = -1 < 0$, so this form is not an inner product. Positive definiteness is what makes $\|v\| = \sqrt{\langle v, v \rangle}$ the length of a vector (Definition 7.15). ◀

Proposition 7.7 (Basic properties). In any inner product space, $\langle v, 0 \rangle = \langle 0, v \rangle = 0$. Moreover, if $\langle u, x \rangle = \langle w, x \rangle$ for all x , then $u = w$.

Proof. By linearity in the second slot, $\langle v, 0 \rangle = \langle v, 0 \cdot 0 \rangle = 0 \cdot \langle v, 0 \rangle = 0$, and by conjugate symmetry $\langle 0, v \rangle = \overline{\langle v, 0 \rangle} = 0$. For the second claim: by conjugate-linearity in the first slot, $\langle u - w, x \rangle = 0$ for all x . Taking $x = u - w$ gives $\langle u - w, u - w \rangle = 0$. By positive definiteness $u - w = 0$, that is, $u = w$. ■

Example 7.8 (Checking the axioms). We check the axioms for the standard product on $\mathbb{C}^n$.

- (1) Conjugate symmetry: $\langle u, v \rangle = \sum_k \overline{u_k} v_k = \overline{\sum_k \overline{v_k} u_k} = \overline{\langle v, u \rangle}$.
- (2) Linearity in the second slot follows from the distributive law for numbers.
- (3) Positive definiteness: $\langle v, v \rangle = \sum_k |v_k|^2 \geq 0$, and this sum is zero only when every coordinate $v_k = 0$.

The inner product of functions from **Example 7.3** is checked in the same way, with the sums replaced by integrals. ◀

Remark 7.9 (Why the conjugation is needed). Why do we need the conjugation in the first slot? Without it the number $\langle v, v \rangle$ could be complex, and the condition $\langle v, v \rangle \geq 0$ would make no sense. We could then not define the length of a vector. Conjugating the first slot keeps the product linear in the second slot and at the same time makes $\langle v, v \rangle$ real and nonnegative.

2 Dirac notation

Physicists write vectors as $|\psi\rangle$ and call them *kets*. To each vector ϕ Dirac assigns a *bra* $\langle\phi|$. It is the linear map that sends each ket $|\psi\rangle$ to a number, the inner product:

$$\langle\phi|: |\psi\rangle \mapsto \langle\phi|\psi\rangle = \langle\phi, \psi\rangle.$$

A linear map from a space V to the scalars is called a *linear functional*. Linear functionals are studied in more detail in Lecture 8.

The inner product is conjugate-linear in the first slot, so the correspondence $|\phi\rangle \mapsto \langle\phi|$ is conjugate-linear too:

$$a|\phi\rangle + b|\chi\rangle \mapsto \overline{a}\langle\phi| + \overline{b}\langle\chi|.$$

So the bracket $\langle\phi|\psi\rangle$ is linear in the ket and conjugate-linear in the bra, exactly as **Definition 7.2** requires.

The product in the opposite order, $|\psi\rangle\langle\phi|$, is no longer a number but an *operator*. It sends a vector $|\chi\rangle$ to the vector

$$|\psi\rangle\langle\phi|\chi\rangle,$$

that is, to a multiple of $|\psi\rangle$. Such an operator is called an *outer product*. Its rank is one, because all its values are multiples of $|\psi\rangle$. In Lecture 8 we will build projections and resolutions of the identity from outer products.

Remark 7.10 (The Born rule). In quantum mechanics the state of a system is described by a vector $|\psi\rangle$ with $\langle\psi|\psi\rangle = 1$. Let $|\phi\rangle$ be another such state. The number $\langle\phi|\psi\rangle$ is called a *probability amplitude*. By the *Born rule*, the square of its modulus, $|\langle\phi|\psi\rangle|^2$, is the probability that a measurement finds the system, prepared in the state $|\psi\rangle$, in the state $|\phi\rangle$. You will study this rule in detail in the Quantum Mechanics course. The Cauchy–Schwarz inequality will show that this probability never exceeds 1.

Example 7.11 (An amplitude). Let

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad \text{and} \quad |\phi\rangle = |0\rangle,$$

where $|0\rangle$ and $|1\rangle$ are two states satisfying $\langle 0|0\rangle = \langle 1|1\rangle = 1$ and $\langle 0|1\rangle = 0$. Then

$$\langle\phi|\psi\rangle = \frac{1}{\sqrt{2}}(\langle 0|0\rangle + \langle 0|1\rangle) = \frac{1}{\sqrt{2}},$$

so the probability of finding the system in the state $|0\rangle$ is $|\langle\phi|\psi\rangle|^2 = \frac{1}{2}$. ◀

Example 7.12 (An outer product as a matrix). In $\mathbb{C}^2$ take $|0\rangle = (1, 0)$ and $|1\rangle = (0, 1)$. We write kets as columns and bras as rows (conjugate transposes). Then the outer product

$$|0\rangle\langle 0| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

is the projection onto the direction of $|0\rangle$:

$$|0\rangle\langle 0|(a|0\rangle + b|1\rangle) = a|0\rangle\langle 0|0\rangle + b|0\rangle\langle 0|1\rangle = a|0\rangle.$$

Adding the two outer products $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$ gives the identity matrix:

$$|0\rangle\langle 0| + |1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

Such an expression for the identity matrix is called a *resolution of the identity*. In the following lectures we will build projections and other operators from outer products like these. ◀

Remark 7.13 (Expansions and expectation values). Let the vectors $|n\rangle$ form an orthonormal basis (Definition 7.15). Then the *completeness relation* $\sum_n |n\rangle\langle n| = I$ holds; we prove it in Lecture 8. Applying it to a vector $|\psi\rangle$ gives the expansion

$$|\psi\rangle = \sum_n |n\rangle\langle n|\psi\rangle,$$

whose coefficients are the amplitudes $\langle n|\psi\rangle$.

In physics a measurable quantity, such as a spin component (Lecture 5), is described by an operator A . Such an operator is called an *observable*. Its *expectation value* in the state $|\psi\rangle$ is the number $\langle A \rangle = \langle\psi|A|\psi\rangle$, the inner product of $|\psi\rangle$ and $A|\psi\rangle$. It is the average of the measured values when the measurement is repeated many times in the same state. We show in Lecture 9 that expectation values of observables are real.

Example 7.14 (An expectation value). Let $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, and let the observable be the third Pauli matrix $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$. It describes the spin component along the z axis, and the possible measured values are its eigenvalues $+1$ and -1 . Since $\sigma_z|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, the expectation value is

$$\langle\sigma_z\rangle = \langle\psi|\sigma_z|\psi\rangle = \frac{1}{2}(\langle 0|0\rangle - \langle 0|1\rangle + \langle 1|0\rangle - \langle 1|1\rangle) = \frac{1}{2}(1 - 1) = 0.$$

The result can also be seen without computation. In Example 7.11 we saw that a measurement finds the system in the state $|0\rangle$ (value $+1$) with probability $\frac{1}{2}$, and in the state $|1\rangle$ (value -1) with probability $\frac{1}{2}$. So the average of the values is zero. ◀

3 Norm, orthogonality, and the Pythagorean theorem

Definition 7.15 (Norm and orthogonality). The *norm* of a vector v is

$$\|v\| = \sqrt{\langle v, v \rangle}.$$

A vector with $\|v\| = 1$ is a *unit vector* (in physics, a *normalized state*). Two vectors are *orthogonal*, $u \perp v$, if $\langle u, v \rangle = 0$. Vectors $e_1, \dots, e_m$ are *orthonormal* if they are unit vectors and pairwise orthogonal, that is,

$$\langle e_i, e_j \rangle = \delta_{ij},$$

where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$ (the *Kronecker delta*). Orthonormal *bases* are studied in detail in Lecture 8.

The norm is *homogeneous*: for every scalar a ,

$$\|av\| = \sqrt{\langle av, av \rangle} = \sqrt{a \bar{a} \langle v, v \rangle} = |a| \|v\|.$$

Moreover, by positive definiteness the norm is zero only when $v = 0$. The number $\|u - v\|$ is called the *distance* between the vectors u and v . Distances and convergence are studied in Lectures 13–14.

The norm and orthogonality are linked by the Pythagorean theorem.

Theorem 7.16 (Pythagorean theorem). If $u \perp v$, then

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

Proof. Expanding the inner product in both slots gives

$$\|u + v\|^2 = \langle u + v, u + v \rangle = \|u\|^2 + \langle u, v \rangle + \langle v, u \rangle + \|v\|^2.$$

When $u \perp v$, both cross terms $\langle u, v \rangle = \langle v, u \rangle = 0$ vanish, and $\|u\|^2 + \|v\|^2$ remains. ■

Example 7.17 (Normalizing a state). The squared norm of the vector $|\psi\rangle = (1, i, 1) \in \mathbb{C}^3$ is

$$\|\psi\|^2 = \langle \psi, \psi \rangle = |1|^2 + |i|^2 + |1|^2 = 3.$$

Dividing the vector by its norm $\sqrt{3}$ gives the normalized state $\frac{1}{\sqrt{3}}(1, i, 1)$. It is a unit vector describing the same physical state as $|\psi\rangle$. ◀

Definition 7.18 (Orthogonal complement). The *orthogonal complement* of a subspace $U \subseteq V$ is the set

$$U^\perp = \{v \in V : \langle u, v \rangle = 0 \text{ for all } u \in U\}.$$

It is also a subspace of V .

We define the orthogonal complement here because only the inner product is needed to define it. We will use it in Lecture 8. There we will split V into the direct sum $U \oplus U^\perp$ and find a formula for the best approximation of a vector in the subspace U .

Proposition 7.19 (Pythagoras for many vectors). If the vectors $v_1, \dots, v_m$ are pairwise orthogonal, then

$$\|v_1 + \dots + v_m\|^2 = \|v_1\|^2 + \dots + \|v_m\|^2.$$

Proof. Expand $\langle \sum_j v_j, \sum_k v_k \rangle = \sum_{j,k} \langle v_j, v_k \rangle$. When $j \neq k$, the term $\langle v_j, v_k \rangle$ is zero by orthogonality. Only the terms with $j = k$ remain, that is, $\sum_j \|v_j\|^2$. ■

Example 7.20 (Pythagoras with three components). The standard basis vectors of $\mathbb{R}^3$ are pairwise orthogonal. The vector

$$v = (2, -1, 2) = 2e_1 - e_2 + 2e_3$$

is a sum of three pairwise orthogonal vectors, so by **Proposition 7.19**

$$\|v\|^2 = 2^2 + (-1)^2 + 2^2 = 9,$$

and $\|v\| = 3$. This is the familiar Euclidean length. The norm of a unit vector in an orthonormal basis is computed in the same way: $\|\psi\|^2 = \sum_n |\langle n | \psi \rangle|^2 = 1$. Physically this says that the probabilities of all measurement outcomes add up to one. ◀

Example 7.21 (Linear independence and the Gram matrix). The *Gram matrix* of vectors $v_1, \dots, v_m$ has entries $G_{jk} = \langle v_j, v_k \rangle$. The vectors are linearly independent if and only if the matrix G is invertible. Indeed, if $\sum_k c_k v_k = 0$, then pairing this equation with each v_j gives $Gc = 0$; the converse also holds. For example, the Gram matrix of the vectors $v_1 = (1, 0)$ and $v_2 = (1, 1)$ in $\mathbb{R}^2$ is $G = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, and $\det G = 1 \neq 0$. So these vectors are linearly independent. $\triangleleft$

Example 7.22 (An orthonormal pair). In $\mathbb{C}^2$ take the vectors

$$e_1 = \frac{1}{\sqrt{2}}(1, 1) \quad \text{and} \quad e_2 = \frac{1}{\sqrt{2}}(1, -1).$$

They satisfy $\langle e_1, e_1 \rangle = \langle e_2, e_2 \rangle = 1$ and $\langle e_1, e_2 \rangle = \frac{1}{2}(1 - 1) = 0$, so they are orthonormal. Every vector $u = (a, b)$ expands as

$$u = \langle e_1, u \rangle e_1 + \langle e_2, u \rangle e_2.$$

Indeed $\langle e_1, u \rangle = \frac{a+b}{\sqrt{2}}$ and $\langle e_2, u \rangle = \frac{a-b}{\sqrt{2}}$, so the right-hand side equals $\frac{a+b}{2}(1, 1) + \frac{a-b}{2}(1, -1) = (a, b)$. Such expansions are studied in general in Lecture 8. $\triangleleft$

Example 7.23 (Norms and inner products of functions). Take the space $\mathcal{C}[0, 1]$ with the inner product $\langle f, g \rangle = \int_0^1 \overline{f} g \, dt$. The squared norm of the function $f(t) = t$ is

$$\|f\|^2 = \int_0^1 t^2 \, dt = \frac{1}{3},$$

so $\|f\| = 1/\sqrt{3}$. The functions 1 and t are not orthogonal, because

$$\langle 1, t \rangle = \int_0^1 t \, dt = \frac{1}{2} \neq 0.$$

In the next lecture the Gram–Schmidt process will replace them with an orthogonal pair, the first Legendre polynomials. $\triangleleft$

4 The Cauchy–Schwarz inequality

Every vector u can be split into two parts: one parallel to a given vector v , the other orthogonal to it (Figure 1). This splitting, together with the Pythagorean theorem, proves the main inequality of this lecture.

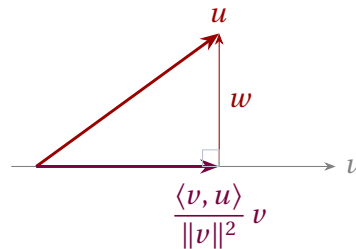


Figure 1: The orthogonal decomposition $u = \frac{\langle v, u \rangle}{\|v\|^2} v + w$ of the vector u , with $w \perp v$. We will apply the Pythagorean theorem to this right triangle.

Theorem 7.24 (Cauchy–Schwarz inequality). In an inner product space, for all u, v ,

$$|\langle u, v \rangle| \leq \|u\| \|v\|.$$

Equality holds if and only if u, v are linearly dependent.

Proof. If $v = 0$, both sides of the inequality are zero, and the vectors u and 0 are linearly dependent. Assume from now on that $v \neq 0$, and write

$$u = \frac{\langle v, u \rangle}{\|v\|^2} v + w, \quad w = u - \frac{\langle v, u \rangle}{\|v\|^2} v.$$

A direct check shows that $w \perp v$ (Figure 1):

$$\langle v, w \rangle = \langle v, u \rangle - \frac{\langle v, u \rangle}{\|v\|^2} \langle v, v \rangle = 0.$$

So the two summands are orthogonal, and by the Pythagorean theorem

$$\|u\|^2 = \frac{|\langle v, u \rangle|^2}{\|v\|^2} + \|w\|^2 \geq \frac{|\langle v, u \rangle|^2}{\|v\|^2}.$$

Multiplying by $\|v\|^2$ and taking square roots gives $|\langle v, u \rangle| \leq \|u\| \|v\|$. By conjugate symmetry $|\langle u, v \rangle| = |\langle v, u \rangle|$, which proves the inequality. Equality holds if and only if $\|w\| = 0$, that is, $w = 0$. This means that u is a multiple of v , that is, the vectors u and v are linearly dependent. ■

Remark 7.25 (The same proof, algebraically [self-study]). Textbooks often give a shorter, purely algebraic proof. For $v \neq 0$, take $\lambda = \langle v, u \rangle / \|v\|^2$ and expand

$$0 \leq \|u - \lambda v\|^2 = \|u\|^2 - \frac{|\langle v, u \rangle|^2}{\|v\|^2}.$$

Rearranging this inequality gives Cauchy–Schwarz at once. It is in fact the same proof: the vector $u - \lambda v$ is the orthogonal part w shown in Figure 1.

Example 7.26 (Cauchy–Schwarz, checked). Take the vectors $u = (1, i)$ and $v = (2, 1 - i)$ from Example 7.4. Their norms and the modulus of their inner product are

$$\|u\| = \sqrt{1+1} = \sqrt{2}, \quad \|v\| = \sqrt{4+2} = \sqrt{6} \quad \text{and} \quad |\langle u, v \rangle| = |1-i| = \sqrt{2}.$$

Then $\sqrt{2} \leq \sqrt{2} \cdot \sqrt{6} = 2\sqrt{3}$. The inequality is strict, because the vectors u and v are linearly independent. ◀

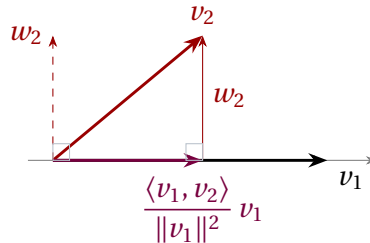


Figure 2: The vector w_2 is obtained from v_2 by subtracting its part parallel to v_1 . The dashed arrow is the same vector w_2 drawn from the origin; it is perpendicular to v_1 .

Example 7.27 (Orthogonalizing a pair). The splitting in the proof of Cauchy–Schwarz gives a simple way to obtain an orthogonal pair. If the vectors v_1 and v_2 are linearly independent, then the vector

$$w_2 = v_2 - \frac{\langle v_1, v_2 \rangle}{\|v_1\|^2} v_1$$

is orthogonal to v_1 (Figure 2). For example, in $\mathbb{R}^2$ take $v_1 = (1, 0)$ and $v_2 = (1, 1)$. Then

$$w_2 = (1, 1) - 1 \cdot (1, 0) = (0, 1).$$

Repeating this step for every basis vector gives the Gram–Schmidt process, which we study in Lecture 8. ◀

Remark 7.28 (Amplitudes are bounded). When $|\phi\rangle$ and $|\psi\rangle$ are unit vectors, Cauchy–Schwarz gives

$$|\langle \phi | \psi \rangle| \leq 1.$$

So the modulus of an amplitude never exceeds one. Hence the Born-rule number $|\langle \phi | \psi \rangle|^2$ lies between zero and one, as a probability must. In Lecture 9 the same inequality will give the Heisenberg uncertainty relation.

Remark 7.29 (Classical inequalities as special cases). Applying Cauchy–Schwarz to the function and Frobenius inner products of Examples 7.3 and 7.5 immediately gives two inequalities:

$$\left| \int_a^b \bar{f} g \, dt \right|^2 \leq \int_a^b |f|^2 \, dt \int_a^b |g|^2 \, dt, \quad |\text{tr}(A^\dagger B)|^2 \leq \text{tr}(A^\dagger A) \text{tr}(B^\dagger B).$$

These inequalities for integrals and matrices are awkward to prove directly. But both are special cases of one general inequality.

5 Triangle inequality, parallelogram law, and polarization

Theorem 7.30 (Triangle inequality). In an inner product space, for all u, v ,

$$\|u + v\| \leq \|u\| + \|v\|.$$

Proof. As in the proof of the Pythagorean theorem, expand $\|u + v\|^2$. The cross terms add up to $\langle u, v \rangle + \langle v, u \rangle = 2 \text{Re} \langle u, v \rangle$, because $\langle v, u \rangle = \overline{\langle u, v \rangle}$. Next we use $\text{Re } z \leq |z|$ and Cauchy–Schwarz:

$$\|u + v\|^2 = \|u\|^2 + 2 \text{Re} \langle u, v \rangle + \|v\|^2 \leq \|u\|^2 + 2|\langle u, v \rangle| + \|v\|^2 \leq (\|u\| + \|v\|)^2.$$

Taking square roots gives the triangle inequality. ■

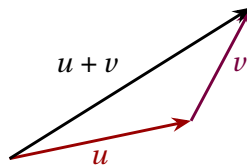


Figure 3: The triangle inequality. The length $\|u + v\|$ of the direct path does not exceed the sum $\|u\| + \|v\|$ of the lengths of the two segments.

Expanding $\|u \pm v\|^2$ gives two more useful identities: the parallelogram law and the polarization identity.

Proposition 7.31 (Parallelogram law and polarization identity). In a complex inner product space, for all u, v , the *parallelogram law*

$$\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2$$

holds. Moreover, the inner product can be recovered from the norm by the *polarization identity*

$$\langle u, v \rangle = \frac{1}{4} \sum_{\alpha \in \{1, -1, i, -i\}} \alpha \|v + \alpha u\|^2.$$

Proof. (1) *Parallelogram law.* Expanding $\|u \pm v\|^2 = \langle u \pm v, u \pm v \rangle$ gives

$$\|u \pm v\|^2 = \|u\|^2 + \|v\|^2 \pm 2\Re\langle u, v \rangle.$$

Adding the two equations (with the plus and with the minus sign) cancels the cross terms and gives the parallelogram law. (2) *Polarization identity.* Each $\alpha \in \{1, -1, i, -i\}$ has $|\alpha| = 1$, so the same expansion gives

$$\|v + \alpha u\|^2 = \|v\|^2 + \|u\|^2 + \alpha \langle v, u \rangle + \bar{\alpha} \langle u, v \rangle.$$

Multiplying by α and summing over all four values of α , we use $\sum \alpha = 0$, $\sum \alpha^2 = 0$ and $\sum \alpha \bar{\alpha} = 4$. What remains is $\sum \alpha \|v + \alpha u\|^2 = 4\langle u, v \rangle$. ■

The polarization identity shows that the inner product is completely determined by the lengths of vectors. In a real space it simplifies to $\langle u, v \rangle = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2)$. In a real space we can also define the *angle* θ between nonzero vectors:

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}.$$

By Cauchy–Schwarz the right-hand side lies in $[-1, 1]$, so such an angle exists.

The parallelogram law helps to recognize whether a given measure of length can come from some inner product. If the law fails, it cannot (**Example 7.35**).

Example 7.32 (An angle and the parallelogram law). In $\mathbb{R}^2$ take the vectors $u = (1, 0)$ and $v = (1, 1)$. They have $\langle u, v \rangle = 1$, $\|u\| = 1$, $\|v\| = \sqrt{2}$, so $\cos \theta = \frac{1}{\sqrt{2}}$ and $\theta = 45^\circ$. Let us also check the parallelogram law:

$$\|u + v\|^2 + \|u - v\|^2 = \|(2, 1)\|^2 + \|(0, -1)\|^2 = 5 + 1 = 6 = 2\|u\|^2 + 2\|v\|^2.$$

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Example 7.33 (Recovering the product from lengths). For the same vectors u and v , the real polarization identity gives

$$\langle u, v \rangle = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2) = \frac{1}{4}(5 - 1) = 1.$$

This agrees with the value computed directly. <

Example 7.34 (Distance and the triangle inequality). Take the vectors $u = (1, i)$ and $v = (2, 1 - i)$ from **Example 7.4**. The distance between them is

$$\|u - v\| = \|(-1, 2i - 1)\| = \sqrt{1 + 5} = \sqrt{6}.$$

The triangle inequality $\sqrt{6} \leq \|u\| + \|v\| = \sqrt{2} + \sqrt{6}$ holds. So in a complex space, too, distances behave as in ordinary geometry. <

Example 7.35 (A norm with no inner product). The length of a vector in the plane can be measured in other ways too, for example by the larger of the moduli of its coordinates:

$$\|(x, y)\|_\infty = \max(|x|, |y|).$$

This measure of length is called the *max norm*. The general notion of a norm is defined in Lecture 13. The max norm on $\mathbb{R}^2$ violates the parallelogram law. Take $u = (1, 0)$ and $v = (0, 1)$. Then

$$\|u + v\|_\infty^2 + \|u - v\|_\infty^2 = 1 + 1 = 2,$$

but $2\|u\|_\infty^2 + 2\|v\|_\infty^2 = 4$. By **Proposition 7.31** every norm defined by an inner product satisfies the parallelogram law. Hence no inner product can induce the norm $\|\cdot\|_\infty$. ◀

Remark 7.36 (The Jordan–von Neumann theorem [self-study]). The converse is also true. If a norm satisfies the parallelogram law, then the right-hand side of the polarization identity defines an inner product, and this inner product induces the norm. This is the *Jordan–von Neumann theorem*. We will not prove it; it comes up again in Lecture 13.

6 States, amplitudes, and what comes next

Let us sum up what the ideas of this lecture mean in quantum mechanics. A physical state is described by a unit vector $|\psi\rangle$. The vectors $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ describe the same state, because the common factor $e^{i\theta}$, called a *phase*, changes no probabilities. The number $|\langle\phi|\psi\rangle|^2$ is the probability of finding the system in the state $|\phi\rangle$. A measurement never confuses orthogonal states ($\langle\phi|\psi\rangle = 0$): a system in the state $|\psi\rangle$ is never found in the state $|\phi\rangle$. In Lecture 9 we prove that eigenvectors of an observable with different eigenvalues are orthogonal. So different measurement outcomes correspond to orthogonal states.

Example 7.37 (Qubit states). A quantum system whose states are the unit vectors of $\mathbb{C}^2$ is called a *qubit*. The vectors $|0\rangle$ and $|1\rangle$ are those of **Example 7.12**. Let

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad \text{and} \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

These vectors are orthonormal. For example, $\langle+|-\rangle = \frac{1}{2}(\langle 0|0\rangle - \langle 1|1\rangle) = 0$, so a system in the state $|+\rangle$ is never found in the state $|-\rangle$. But the inner product of each of them with $|0\rangle$ equals $\frac{1}{\sqrt{2}}$, as in **Example 7.11**. So a measurement in the basis $|0\rangle, |1\rangle$ gives each outcome with probability $\frac{1}{2}$ in both cases. Such a measurement therefore does not tell $|+\rangle$ and $|-\rangle$ apart, although they differ only in the sign in front of $|1\rangle$. A measurement in the basis $|+\rangle, |-\rangle$ does tell them apart. ◀

Remark 7.38 (Toward Hilbert space [self-study]). In quantum mechanics the state space is called a *Hilbert space*. It is a complex inner product space with one more property, *completeness*: every sequence of vectors whose terms get arbitrarily close to each other has a limit in the same space. Every finite-dimensional inner product space is complete, so the difference is invisible in our lectures. For infinite-dimensional spaces, such as function spaces, completeness is an extra requirement. It is studied in Lecture 13.

Example 7.39 (The closest point on a line). Let e be a unit vector and u any vector. Which multiple ce of e is closest to u ? The error $u - \langle e, u \rangle e$ is orthogonal to e , as in the proof of Cauchy–Schwarz. The vector $u - ce$ is the sum of this error and a multiple of e , so by the Pythagorean theorem

$$\|u - ce\|^2 = \|u - \langle e, u \rangle e\|^2 + |c - \langle e, u \rangle|^2.$$

This expression is smallest when $c = \langle e, u \rangle$. So the closest vector is $\langle e, u \rangle e$, the *orthogonal projection* of u onto the line in the direction of e . The least-squares method and the orthonormal expansions of Lecture 8 rest on this idea. ◀

In the next lecture we answer the question: does every inner product space have a basis of pairwise orthogonal unit vectors, that is, an *orthonormal basis*? In finite dimensions the

answer is yes: the Gram–Schmidt process constructs one. In an orthonormal basis the inner product becomes the familiar sum $\sum_k \overline{u_k} v_k$, and the matrices of operators become especially simple.

Summary of main concepts

An inner product is conjugate symmetric, positive definite, and linear in the second slot. Hence it is conjugate-linear in the first slot, that is, sesquilinear (the physics convention). In Dirac notation $\langle \phi | \psi \rangle$ is linear in the ket and conjugate-linear in the bra. The inner product defines the norm $\|v\| = \sqrt{\langle v, v \rangle}$ and orthogonality $\langle u, v \rangle = 0$, and the Pythagorean theorem links them. The Cauchy–Schwarz inequality $|\langle u, v \rangle| \leq \|u\| \|v\|$ follows from the orthogonal decomposition and the Pythagorean theorem. The triangle inequality follows from it. The parallelogram law and the polarization identity show that the inner product is determined by the lengths of vectors. In quantum mechanics states are unit vectors, $|\langle \phi | \psi \rangle|^2$ is a probability, and a measurement always tells orthogonal states apart. These ideas are needed for the spectral theorem in Lecture 10.

- ▶ **Inner product, physics convention** — conjugate symmetric, linear in the second slot, positive definite; $\langle \phi | \psi \rangle$ is linear in the ket.
- ▶ **Norm and orthogonality** — $\|v\| = \sqrt{\langle v, v \rangle}$; $\langle u, v \rangle = 0$; the Pythagorean theorem.
- ▶ **Cauchy–Schwarz inequality** — $|\langle u, v \rangle| \leq \|u\| \|v\|$; the modulus of an amplitude is at most one.
- ▶ **Triangle inequality, parallelogram law, and polarization** — properties of lengths and distances; $\langle \cdot, \cdot \rangle$ is recovered from $\|\cdot\|$.
- ▶ **Physical picture** — states are unit vectors, $|\langle \phi | \psi \rangle|^2$ is a probability, orthogonality means distinguishability.